Solicitation Number: USSOCOM RFI REPTILE (Rapid Experimentation of Prototyping Technology in Learning Environments): Counter-Unmanned-Systems (C-UxS)

Notice Type: Special Notice

TYPE: A–Research and Development

NAICS: 541715 Research and Development in the Physical, Engineering, and Life Sciences (except Nanotechnology and Biotechnology)

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A. INTRODUCTION: REPTILE (Rapid Experimentation of Prototyping Technology in Learning Environments)

This Request for Information (RFI) is NOT a solicitation for proposals, proposal abstracts, or quotations. The purpose of this RFI is to solicit technology experimentation candidates from Research and Development (R&D) organizations, private industry, and academia for inclusion in future experimentation events coordinated by the U. S. Special Operations Command (USSOCOM). USSOCOM invites industry, academia, individuals, and Government labs to submit technology experimentation nominations addressing innovative technologies identified in paragraph 4 below. The intent is to provide participants with the opportunity to gain Special Operations Forces (SOF) insight/perspective on participant technologies.

The REPTILE event will explore emerging technologies, technical applications, and their potential to provide solutions for future SOF capabilities. This RFI is for REPTILE 26. This event will occur over the period of one week.

Dates/Theme/Location:
June 2026: Counter-Unmanned-Systems (C-UxS) – Gulfport, MS.

B. OBJECTIVE:

1. The REPTILE event provides an opportunity for technology developers to operate and have their technology assessed by operational personnel through environmental scenarios to determine how their technology development efforts and ideas may

support or enhance SOF capability needs. The environment facilitates a collaborative relationship between Government, academia, and industry to promote the identification and assessment of emerging technologies.

2. The deadline for nomination package(s) is **April 21, 2026, at 12:00 Noon EDT.** After reviewing the nomination submissions, the Government may invite select candidates to demonstrate their technologies at the USSOCOM sponsored REPTILE 2026 event. Experiments will be conducted during June 2026, Gulf Coast area, and will explore emerging technology solutions and revolutionary improvements in relevant technologies. Preference will be given to materiel solutions between a Technology Readiness Level (TRL) of 6 and above. There is no intention on the part of USSOCOM to purchase or procure equipment based solely on participation in REPTILE 2026.

3. **Experimentation Focus:** The primary intent of this event is to highlight technologies that support C-UxS and friendly advanced Unmanned Systems (UxS) operations. Experiments will be conducted in unimproved expeditionary-like conditions with a dismounted special operations team. Relevant networks and an Android Tactical Assault Kit (ATAK) capability will be available at the event for integration and/or operations with participating experiments. Depending on the technology selection some integration and/or collaboration with existing SOF technology may be an opportunity.

4. **Experimentation Problem Statement:** Special Operators have a need to defeat a swarm of up to 100x, Group 1-2, autonomous drones while operating in a denied areas from a static position within wooded terrain in day and night conditions.

5. **Design Reference Mission:** A special operations team infiltrated behind enemy lines to conduct a 5-day special reconnaissance mission. The team established a Mission Support Site and Observation Point in wooded terrain observing and reporting on an enemy facility. The team emplaced detection technologies and set up defensive measures to counter drone swam threats. After 48 hours, the team identified indications, warnings and their position was compromised and the enemy was sending a drone swarm to find, fix, and finish the team. The team's detection equipment identified the threat and relayed it to the team and its defensive effectors. The effectors engaged with and destroyed the threat, allowing the team to successfully exfil without any casualties.

6. Technology areas to explore during the event include the following:

6.1. **Counter-Unmanned Systems (C-UxS) Technologies:** USSOCOM is seeking C-UxS technologies that can quickly detect, track, identify, classify, and neutralize individual Unmanned Systems (UxS) and swarms of UxS. Technologies will operate in multi-domains including ground, air, maritime and at times may traverse multiple in one operation. Key areas of interest include:

6.1.1. Integrated detection and engagement capabilities.

6.1.2 Early warning systems.

6.1.3 Semi-autonomous and autonomous platforms.

6.1.4 Integration into existing C2 systems.

6.1.5 Electromagnetic (EM) hardened systems.

6.1.6 Low Collateral Effectors: Precision non-kinetic and kinetic solutions for use in complex environments.

6.1.7 Electromagnetic protection (EP) systems that can detect, characterize, and geolocate electromagnetic interference (EMI).

6.1.8. EMI detection finding (direction, target location- within 40m).

6.2. Modular, Low SWaP- CUxS & Rapidly Produced (All UxS Types):

6.2.1. Modular components, multi-factor components and integrations.

6.2.2. Low-cost/attritable/disposable/rapidly produced CUxS effectors.

6.2.3. Software-agnostic.

6.2.4. Kinetic/non-kinetic and or mixed modular payloads.

6.2.5. Edge UxS production and maintenance.

6.2.6. Tactical UxS weaponry.

6.2.7. Low SWaP jam-resistant GPS antennas or other operational PNT solutions.

6.3. Counter-UxS Capability Focus:

6.3.1. Operator-worn/portable UxS detection/localization/classification (Passive).

6.3.2. Novel defeat capabilities (portable, UxS-mounted, vehicle-mounted low SWaP).

6.3.3. Low-SWaP kinetic defeat (not ammunition specifically)

6.3.4. Low-collateral kinetic

6.3.5. Omni-detection coverage.

- 6.3.6. Allows short-to-medium term stationary deployment and/or mobile options for tactical teams.
- 6.3.7. RF Based cyber takeover technology.
- 6.3.8. Provides end-to-end detection and mitigation for situational awareness and operational continuity.

6.4. Unmanned and Autonomous Systems (UAS/UxS) Technologies. We are seeking the best Group 1-2 UAS technologies that can operate despite of adversarial C-UxS technologies. Desired capabilities include:

- 6.4.1. EM Hardening: Robust C2 links and GPS/GNSS receivers.
- 6.4.2. Swarming/Collaborative Autonomy: Multiple UxS operating as a cohesive unit to survive threats.
- 6.4.3. Tethered Systems (with mobility).
- 6.4.4. Enhanced Survivability: Low-observable characteristics and onboard threat detection.
- 6.4.5. Multi-purpose, multi-domain technologies and/or systems.
- 6.4.6. Deployable from undersea.
- 6.4.7. Rapid deployment to underwater location.
- 6.4.8. Vertical integration.
  - 6.4.8.1. Ability to deliver payloads to undersea locations.
- 6.4.9. Payload retrieval from undersea.
- 6.4.10. Waterproof, robust, weatherproof platform.
- 6.4.11. Fast transition from air to sea, or sea to air.

6.5. Unmanned Systems (UxS) - All Domains

- 6.5.1. Multi-Modal, Multi-Domain Capabilities:
  - 6.5.1.1. Single systems (air, land, sea, space).
  - 6.5.1.2. Low-cost, attritable sensors/payloads w/ volatile firmware & remote zeroizing (SIGINT, EW, IMINT, GEOINT, ELINT, MASINT), including RF deception/decoy payloads.
  - 6.5.1.3. Low SWaP phased antenna arrays.
  - 6.5.1.4. Multi-modal, cognitive, intelligent communication systems
  - 6.5.1.5. Biomimicry. Modeled after living organisms.
  - 6.5.1.6. Multi-domain Automatic Threat/Target Recognition (ATR) algorithms (agnostic datasets).
  - 6.5.1.7. No-code/low-code multi-domain edge ATR training.

Proposed solutions that include components enhancing resilience and survivability of SOF teams in Counter UxS operations may include the following components and/or technologies intended to be evaluated or assessed.

6.6. Heterogeneous Teaming & UxS Swarming:

6.6.1. UxS common controllers/architectures.
6.6.2. Inter-agent communications.
6.6.3. AI-enabled swarming algorithms.
6.6.4. Multi-domain delivery/charging platforms.
6.6.5. Manned/unmanned teaming with collaborative alerts for situational awareness.

6.7. UxS Human-Machine Interface (HMI) & AI/Sensor Integration:

6.7.1. AI for autonomy.
6.7.2. Novel RF/non-RF control architectures.
6.7.3. Multi-modal UxS control (single/multi-agent, End User Device-based).
6.7.4. Bi-directional state sharing.
6.7.5. Advanced UxS/AR displays.
6.7.6. Hands-free command.
6.7.7. AI-enabled COP.
6.7.8. Multi-sensor fusion.
6.7.9. ATR integration.
6.7.10. Commercial Internet of Things integration.

6.8. Blue Force Awareness Technology:

6.8.1. Devices/tech to distinguish friend/foe on UxS platforms visually/digitally.
6.8.2. Lightweight Air Launched Effects Delivery System.
6.8.3. Lightweight, precision, glide systems.
6.8.4. GPS-denied navigation.
6.8.5. Semi-active laser designator.
6.8.6. Payload <10lbs, overall system <30lbs.

7. **Security/Classification Requirements:** Respondents shall not submit classified information in the technology experimentation nominations. If respondents believe classified information is required for their proposal please contact tech_exp@socom.mil to coordinate appropriate information.

8. **Safety Requirements:**
   8.1. All respondents shall review the TE Safety Guide (**RFI Notice Attachment 3**).

   8.2. Those respondents who are invited to demonstrate their technologies will complete a Deliberate Risk Assessment Worksheet (Department of Defense Form 2977) (**RFI Notice Attachment 4**) in accordance with MIL-STD-882E and the Department of the Army Techniques Publication No. 5-19 (ATP 5-19).

      8.2.1. Risk assessments shall be emailed directly to the **tech_exp@SOCOM.mil** by **01 June 2026**. Respondents should include instructions that describe the safe operation of the device nominated for the experiment.

9. **Frequency Requirements:** If your experiment has **any** device radiating on a given radio frequency or frequency band, you must have prior approval to transmit on that frequency. Prior approval includes:

   9.1. Compliance with Federal Communications Commission (FCC) Title 47, Part 15, or a Special Temporary Authority (STA) from the FCC.

   9.2. If your experiment includes Government-owned equipment, and you will be operating within a Federal Band, you MUST have National Telecommunications and Information Administration (NTIA) frequency approval. Providing JF12 information will advance coordination and approvals as needed.

   9.3. **Respondents are advised NOT to wait for confirmation of selection/invitation to the event before requesting a STA, if needed, from the FCC as such requests can take up to 60 days to get FCC approval**.

   9.4. Your authority to radiate must be emailed directly to tech_exp@socom.mil by **01 June 2026**. All frequency questions shall be directed to the USSOCOM Technical Experimentation team at the email above.

10. **Human Subjects.** It is not anticipated that activities being conducted in this TE event will require the use of research or experimentation involving human subjects. Technology experiment submissions will be reviewed for potential research or experimentation involving human subjects. Any submission that is determined to potentially include research or experimentation involving human subjects will be required to adhere to DoD Instruction 3216.02 “Protection of Human Subjects and Adherence to Ethical Standards in DoD Supported Research” and ensure appropriate Institutional Review Board and DoD Human Research Protections Office approvals prior to conducting those activities.

10.1. Other Special Requirements:

10.2. **DO NOT** submit proposals or marketing demonstrations.

10.3. SUBMIT TECHNOLOGY EXPERIMENTATION NOMINATIONS ONLY.

10.4. EXPERIMENTATION NOMINATION SUBMITTALS FOR THIS RFI WILL ONLY BE ACCEPTED UNTIL THE CLOSING DATE OF **April 21, 2026, 12:00 Noon EDT.**

10.5. No contracts will be awarded based solely on this announcement or any subsequent supplemental RFI announcements. All responses must be in accordance with the topic of this announcement; solutions outside this scope will be excluded.

C. SUBMISSION INSTRUCTIONS:

1. READ ALL OF SECTION C BEFORE ATTEMPTING TO SUBMIT AS THE REQUIREMENTS MAY HAVE CHANGED.

2. Technology experimentation nominations shall be submitted electronically by creating a Scout Card in the Vulcan system at vulcan-sof.com by the respective deadline. Scout Cards must be created within the Call for REPTILE 26: Counter-Unmanned-Systems (C-UxS). The Vulcan system is best accessed by using Firefox or Google Chrome browsers. Associated technology experiments with distinctly different uses or applications should have a separate nomination submitted by each respondent. USSOCOM personnel will review submissions to determine whether an experiment submission will be accepted for invitation to attend the REPTILE event.

3. **CRITICAL**: It takes 35-40 minutes to complete the Scout Card and the Electronic Supplemental Information link. If you try to submit in the last 30 minutes of the day of the Call closing, you will not succeed. **PLAN ACCORDINGLY**! When the Call closes it WILL NOT BE REOPENED.

4. **A complete submission consists of**:
   4.1 Creating a Scout Card in the Vulcan system as described above. **Submitters will need their Scout Card ID number to complete the following step**.

4.2 **DO NOT** attempt this step if you don't have or know the Vulcan Scout Card ID. Completely and Correctly filling in the Supplemental Information Electronic form via https://forms.osi.apps.mil/r/YTkmqLL5D8. This is listed as the last question in the Vulcan submission for this event. The Electronic Supplemental Information form is a **MANDATORY** requirement for **ALL** respondents. Failure to fill out the electronic form, **completely and correctly**, via the link provided is grounds for immediate rejection for this event without further notice.

4.3 Devices with **ANY** radio frequency emissions will state the exact intended frequency or frequencies used by the device(s). **DO NOT** enter state in the electronic form that it will be provided on request. The form is the request.

4.4 An FCC STA, Experimental License, or NTIA document (for developmental radio frequency emitting devices). If neither is available at the time of submission, provide status of your FCC/NTIA request. Those using only the ISM Band must state the **exact frequency range** and must transmit at less than 1watt.

4.4.1 ISM band rules require devices to operate without causing harmful interference and to accept any interference from other devices. In the United States, these rules are governed by the FCC, with Part 18 for ISM equipment and Part 15 for unlicensed communication devices that share ISM frequencies.

4.5 Instructions on how to safely use the technology (as needed).

4.6 If applicable, a picture of the device with a short description of the size (shows the dimensions or places the device next to a ruler, currency, or man-sized object for comparison).

4.7 A signed Photo Release (RFI Attachment 2).

- File named as: **Scout Card ID_Company Name**.
- Example: 99875_AcmeAnvil

5. Selected respondents will be invited to participate in USSOCOM experiments. USSOCOM shall provide venues, supporting infrastructure, and assessment

(operational and technical, based on availability of resources and written request as discussed above) personnel at no cost to invited respondent(s).

5.1. **All respondents' submission costs, travel costs, technology experiments, and experimentation associated costs will be at the respondents' expense**.

5.2. The TE venue will only provide basic access to training areas or ranges (if approved and applicable) to conduct experiments, a facility to connect to the internet, basic venue infrastructure including frequency coordination/deconfliction, and shore power. Invited respondents must be prepared to be self-sufficient during the execution of their experiments and not dependent on venue resources. Time and space will be made available for technology developers to conduct real-time modifications and updates to technologies.

5.3. Technology developers are advised to bring all tools and equipment necessary to present/operate their technology at the event.

D. BASIS FOR SELECTION TO PARTICIPATE: Selection of respondents to participate shall be based on the extent to which the technology represents a potential capability increase to SOF.

Other considerations include:

- Technical maturity
- Relevance of or adaptability to military operations/missions
- Relevance to current operational needs
- Relevance to Event Focus Area

E. ADDITIONAL INFORMATION:

- All efforts shall be made to protect proprietary information that is clearly marked in writing. Technology Developers shall not display any proprietary or classified information at the event. Lessons learned by USSOCOM from these experiments may be broadly disseminated, but only within the Government.

- If selected for participation in Technical Experimentation, developers may be requested to provide additional information that will be used in preparation for the experiments.

F. USE OF INFORMATION: The purpose of this notice is to gain information leading to Government/Industry collaboration for development of USSOCOM technology capabilities and to assist in accelerating the delivery of these capabilities to the SOF warrior. All proprietary information contained in the submission and technology experimentation shall be appropriately marked. The Government will not use proprietary information submitted from any one firm to establish future capability and requirements. During the event USSOCOM may broadcast technology experiments via video teleconference back to USSOCOM and/or their operational components. Respondents to this RFI shall review and submit the photography approval form at **Attachment 2**.

G. SPECIAL NOTICE:

1. Federally Funded Research and Development Centers (FFRDCs) or contractor consultant/advisors to the Government will review and provide support during evaluation of submittals. When appropriate, non-Government advisors may be used to objectively review a particular functional area and provide comments and recommendations to the Government. All advisors shall comply with procurement integrity laws and shall sign non-disclosure statements. The Government shall take into consideration requirements for avoiding conflicts of interest and ensure advisors comply with safeguarding proprietary data. Submission in response to this RFI constitutes approval to release the submittal to approved Government support contractors.

2. There will be foreign attendees (military, government, or industry partners), who are interested in the capabilities being demonstrated at the TE event. Respondents are ultimately responsible for complying with all International Trafficking in Arms Regulations (ITAR) /Export Administration Regulations (EAR) requirements associated with their equipment.

2.1 Before declaring their ITAR/EAR status, respondents to this RFI should verify the ITAR/EAR status of their technology by visiting the **US Department of State Directorate of Defense Trade Controls** (ITAR) website at: (https://www.pmddtc.state.gov/ddtc_public) and the **US Department of Commerce Bureau of Industry and Security** website (https://www.bis.doc.gov/) for more information on Export Administration Regulations (EAR).

2.2 USSOCOM event organizers will restrict access as necessary to assist in protecting ITAR/EAR related technology demonstrations.

H. Per Federal Acquisition Regulation (FAR) 52.215-3, Request for Information or Solicitation for Planning Purposes (Oct 1997):

1. The Government does not intend to award a contract based on this RFI notice or to pay for the information.
2. Although "proposal" and "respondent" are used in this RFI, your responses will be treated as information only. It shall not be used as a proposal.
3. In accordance with FAR 15.209(c), the purpose of this RFI is to solicit technology experimentation candidates from research and development organizations, private industry, and academia for inclusion in future experimentation events coordinated by USSOCOM.

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